



Q.1 Write a program that takes a list of numbers and prints all prime numbers in the list using loops. Do not use any built-in prime-checking functions.

```
In [ ]: numbers=[1,2,3,4,5,6,7]

print("prime numbers in the list are:")

for num in numbers:
    if num>1:
        is_prime=True

        for i in range(2,num):
            if num % i ==0:
                is_prime=False
                break

        if is_prime:
            print(num)
```

```
prime numbers in the list are:
2
3
5
7
```

Q.2 Write a program that generates a multiplication table (1-10) in matrix format using nested loops.

```
In [2]: for i in range(1,11):
        for j in range(1,11):
            print(i * j,end="\t")

        print()
```

1	2	3	4	5	6	7	8	9
10								
2	4	6	8	10	12	14	16	18
20								
3	6	9	12	15	18	21	24	27
30								
4	8	12	16	20	24	28	32	3
6	40							
5	10	15	20	25	30	35	40	
45	50							
6	12	18	24	30	36	42	48	
54	60							
7	14	21	28	35	42	49	56	
63	70							
8	16	24	32	40	48	56	64	
72	80							
9	18	27	36	45	54	63	72	
81	90							
10	20	30	40	50	60	70	80	
90	100							

Q.3 Write a user-defined function that accepts a list of integers and returns:

- Count of even numbers
- Sum of odd numbers
- Use loops and conditional statements inside the function

```
In [7]: def count_even_sum_odd(numbers):
        even_count=0
        odd_sum=0

        for num in numbers:
            if num % 2 ==0:
                even_count +=1
            else:
                odd_sum +=num

        return even_count,odd_sum

numbers=[2,3,5,6,8,9,11,13,15]

even_count,odd_sum=count_even_sum_odd(numbers)

print("Count of even numbers:",even_count)
print("Sum of odd numbers :",odd_sum)
```

Count of even numbers: 1
Sum of odd numbers : 0

Q.4 Create a function that calculates simple interest. Use default arguments for rate and time. Demonstrate calling the function using both positional and keyword arguments.

```
In [1]: def simple_interest(p,r=5,t=2):
        return(p*r*t)/100

        # positional argument
        print(simple_interest(p=1000))

        # keyword argument
        print(simple_interest(p=1000,r=10,t=7))
```

100.0
700.0

Q.5 Create a Student class with attributes: name, marks. Add a method to calculate grade based on marks.

```
In [8]: class Student:
        def __init__(self,name,marks):
            self.name=name
            self.marks=marks

        def grade(self):
            if self.marks>=95:
                return "A++"
            elif self.marks >=85:
                return "A"
            elif self.marks>=75:
                return "B"
            elif self.marks>=65:
                return "C"
            elif self.marks>=45:
                return "D"
            else:
                return "fail"

        s=Student("Shakshi",98)

        print("Name:",s.name)
        print("Marks:",s.marks)
        print("Grade:",s.grade())
```

Name: Shakshi
Marks: 98
Grade: A++

Q.6 Create at least two objects and display their grades.

```
In [11]: s1=Student("Mannu",85)
        s2=Student("Priya",88)
        print(s1.name,"-",s1.grade())
        print(s2.name,"-",s2.grade())
```

Mannu - A
Priya - A

Q.7 Create a BankAccount class with private attributes for balance. Provide methods to deposit and withdraw money. Ensure balance cannot be accessed directly.

```
In [2]: class BankAccount:
        def __init__(self, balance):
            self.__balance = balance

        def deposit(self, amount):
            self.__balance += amount

        def withdraw(self, amount):
            self.__balance -= amount

        def show(self):
            print(self.__balance)

a = BankAccount(2000)
a.deposit(5000)
a.withdraw(100)
a.show()
```

6900

Q.8 Write a program that takes two numbers from the user and performs division.

- Handle the following exceptions:
- Division by zero
- Invalid input type

```
In [4]: try:
        num1 = int(input("Enter a number:"))
        num2 = int(input("Enter a number:"))

        division = num1 / num2
        print(division)

    except ZeroDivisionError as e:
        print("Cannot divide by zero")

    except ValueError:
        print("Invalid input")
```

Cannot divide by zero

Q.9 Write a program that:

- Creates a file
- Writes student data (name and marks)
- Reads and displays the content

```
In [1]: f = open("student.txt", "w")
        f.write("Name: Sakshi\nMarks: 85")
```

```
f.close()

f = open("student.txt", "r")
print(f.read())
f.close()
```

Name: Sakshi

Marks: 85

Q.10 Write a program that reads numbers from a file and calculates:

- Total
- Average
- Handle the case if the file does not exist

```
In [2]: try:
        f = open("numbers.txt", "r")

        nums = list(map(int, f.read().split()))

        print("Total =", sum(nums))
        print("Average =", sum(nums) / len(nums))

        f.close()

    except FileNotFoundError:
        print("File does not exist")
```

File does not exist